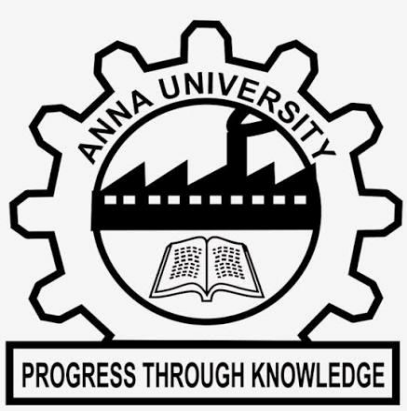
DESIGN AND DEVELOPMENT OF A WEB-BASED RAILWAY TICKET RESERVATION SYSTEM

A SUMMER INTERNSHIP REPORT

Submitted by

R. MADHUMITHA - 810023205064



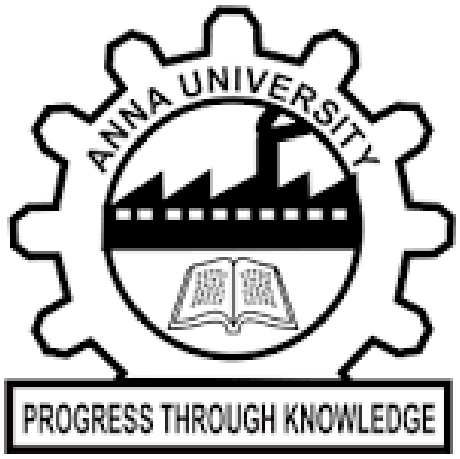
DEPARTMENT OF INFORMATION TECHNOLOGY

UNIVERSITY COLLEGE OF ENGINEERING – (BIT CAMPUS)

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DEPARTMENT OF INFORMATION TECHNOLOGY



BONAFIDE CERTIFICATE

This is to certify this Summer Internship project report **"Design and Development of a Web Based Railway Ticket Reservation System"** is the bonafide work of **R. MADHUMITHA (810023205064)** and has satisfactorily completed the course of **IT3711 - SUMMER INTERNSHIP** for **SEVENTH** semester, **B.Tech. INFORMATION TECHNOLOGY** during the academic year 2026-2027**.**

Faculty in charge **Head of the department**

Submitted for the practical examination held on_____

ABSTRACT

The Design and Development of a Web-Based Railway Ticket Reservation System is a web application developed to simplify and modernize the process of railway ticket reservation. The system provides a convenient platform for passengers to search for trains, check seat availability, select seats, book tickets, make payments, and view or cancel their reservations through an online interface. The main objective of the system is to reduce the time and effort involved in traditional ticket booking while providing an efficient, user-friendly, and reliable reservation experience.

The system is designed with separate functionalities for passengers, clerks, and administrators. Passengers can register and log in to the system, search trains based on source, destination, and journey date, view train schedules and available seats, select their preferred seats, and complete the booking process. After successful booking, the system generates a ticket containing important details such as the passenger information, train number, journey details, seat number, and PNR number. Passengers can also view their booking status and cancel tickets when required. Clerks can assist passengers in making reservations and managing bookings, while administrators can manage train details, users, schedules, and reservation records.

The application follows a structured web-based architecture consisting of a frontend, backend, and database. The frontend provides an interactive and responsive user interface, while the backend handles business logic, authentication, reservation processing, and communication with the database. The database stores passenger details, train information, seat availability, bookings, payments, and other relevant records securely. Technologies such as HTML, CSS, JavaScript, React.js, Node.js, Express.js, and MySQL can be used to implement the system.

The proposed system aims to provide faster, more accessible, and efficient railway ticket reservation services while reducing manual work and improving the accuracy of reservation management. Overall, the project demonstrates the practical application of web development technologies in developing a real-world railway reservation system.

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1. INTRODUCTION

The primary objective of this project is to design and develop a user-friendly web-based railway ticket reservation system that simplifies the process of railway ticket booking. The system aims to provide passengers with an easy and convenient platform to access railway reservation services through a web browser.

The system aims to enable users to search for trains, check seat availability, enter passenger details, and book tickets online**.** It also provides facilities for payment processing, ticket cancellation, and checking PNR and booking status, thereby reducing the need for manual reservation procedures.

Another objective is to efficiently manage railway-related information through a centralized database**.** Passenger details, train information, seat availability, booking records, and payment details can be stored and retrieved systematically, helping to improve data accuracy and reduce booking errors.

The project also aims to provide an administrator module for managing trains, schedules, users, seats, and booking information. Secure authentication and appropriate access control are incorporated to protect user information and ensure that administrative functions are accessible only to authorized users.

Finally, the project aims to develop a responsive, reliable, and efficient web application while providing practical experience in front-end development, back-end programming, database integration, testing, and debugging. The system is intended to reduce processing time and improve the overall convenience and efficiency of railway ticket reservation.

1.1 OBJECTIVE

The major objectives of the project are:

- To design and develop a user-friendly web-based railway ticket reservation system.
- To provide online train searching and seat availability checking.
- To enable passengers to book and cancel railway tickets conveniently.

- To maintain passenger, train, booking, and payment information efficiently.
- To implement secure user authentication and role-based access.
- To generate digital tickets and PNR details after successful booking.
- To provide an administrative interface for managing railway-related information.
- To ensure accurate and reliable booking information through proper database management.
- To perform system testing and debugging to ensure reliable application performance.
- To reduce manual effort and improve the overall efficiency and user experience of railway ticket reservation.

SPECIFIC OBJECTIVES

- To develop a user-friendly and responsive railway reservation website.
- To provide secure user registration and login.
- To enable users to search trains and check seat availability.
- To facilitate online ticket booking, payment, and cancellation.
- To generate digital tickets and PNR details.
- To provide an admin module for managing trains, users, and bookings.
- To maintain passenger and booking data using a centralized database.
- To test the system for functionality, security, and performance.
- To gain practical knowledge of web development and database integration.

1.2 MOTIVATION

The main motivation for developing the Web-Based Railway Ticket Reservation System is to make railway ticket booking faster, easier, and more convenient for passengers. Traditional reservation methods may involve waiting in queues and manual procedures, which can consume time and lead to errors. This project provides an online platform for users to search trains, check seat availability, book and cancel tickets, make payments, and view PNR details. It also helps administrators manage train, passenger, and booking information efficiently. Through this project, practical knowledge of web development, database management, authentication, system integration, and testing can be gained while developing a useful real-world application.

2. COMPANY PROFILE

ADDRESS

Fantasy Solution

Fantasy Solution,

No 16, Samnath Plazza,

Third Floor, Near BG Naidu Sweets,

Melapudur,
Trichy-1.

Contact: 9043095535

Landline: 0431-4971630

Website: www.fantasysolution.in

Company profile

Fantasy solution as a leading IT solution and service provider, provides innovative information technology - enabled solutions and services to meet the demands arising from social transformation, shaping new life styles for individuals and creating values for the society.

Focusing on software technology, Fantasy solution provides industry solutions and product engineering solutions, related software products & platforms, and services, through

seamless integration of software and services, software and manufacturing, as well as technology and industrial management capacity.

Fantasy solution helps industry customers establish best practices in business development and management. The fantasy solution serves include real time projects, web designing, web hosting, software development and training etc, in many of which, has a leading market share. Notably, Fantasy Solution has participated in the formulation of many national IT standards and specifications.

Fantasy solution has the world's leading product engineering capabilities, ranging from consultation, design, R&D, and integration to testing of embedded software, in the fields of automotive electronics, smart devices, digital home products, and IT products. The software provided by fantasy solution runs in a number of globally renowned brands.

Particularly offering the services that include application development & maintenance, ERP implementation & consulting, testing, performance engineering, software localization & globalization, IT infrastructures, BPO, IT education & training, etc.Sticking to its business philosophy and brand commitment of "Beyond Technology", fantasy solution is dedicated to providing innovative information technologies to drive the sustainable development of society, as well as becoming a company that is well recognized and respected by employees, shareholders, customers, and society.

Our services

In this ever-changing environment, keeping a competitive edge means being able to anticipate and respond quickly to changing business conditions. Fantasy solution is a global software development company providing IT solutions to enterprises worldwide. Combining proven expertise in technology, and an understanding of emerging business trends, Fantasy delivers a range of software development solutions that includes e-business solutions, computer telephony, enterprise applications, professional web site design and development, product engineering, Electronic Health Records, CMS Software's, Payment Gateway solutions, Time and attendance tracking software's, Debt collection software's, Appointment Reminder Solutions, Medical Transcription Services etc. We study, design, develop, enhance, customize, implement, maintain and support various aspects of information technology.

We are a professionally recognized software development company having huge experience in developing custom software development and application development best

match to your need and requirements. We have expertise in working with a variety of customers from companies to individuals. Our successful assignments with client companies have established our reputation as superior providers of IT solutions. Services offered by Fantasy includes Software development in .NET, PHP and Java, Web designing & development, Mobile application development in Android platforms, MATLAB for image processing, Network Simulator (NS2), Data mining tools, Big data development using R tool and Python. We aim to carve a position in the forefront, and it is our continuing goal to gain the trust of our clients. Our Motto is to serve the purpose of our clients with perfection.

Mission

Fantasy Solutions' mission includes:

1. Providing high quality software development services, professional consulting and development outsourcing that would improve our customers' operations;
2. Making access to information easier and securer (Enterprise Business);
3. Improving communication and data exchange (Business to Business);
4. Providing our customers with a Value for Money and
5. Providing our employees with meaningful work and advancement opportunities.

Vision

Fantasy Solutions is a leading IT company for Consulting Services and Deployment of best of breed Business Solutions to top tier domestic and international customers.

3. DAILY WORK REPORT

DAY 1

On the first day, I gained knowledge about web development and its importance for creating effective websites. I researched JavaScript, HTML, CSS, and other essential elements of web pages. Additionally, I learned how to set up an environment for development.

DAY 2

I learnt the foundations of HTML (Hypertext Mark-up Language) on the second day. I learned how to arrange material for the web using HTML elements including headings, paragraphs, lists, and links. At the end of the day, I created a simple HTML page that I could view in a web browser.

DAY 3

On the third day, I studied CSS (Cascading Style Sheets) and its use in web design. I learned how to apply CSS styles to enhance the appearance and feel of HTML elements as well as selectors, attributes, and values in CSS.

DAY 4

On the fourth day, I spent more time studying CSS and got to know the CSS box model. I knew how each element's padding, borders, margins, and content were arranged. I've researched a number of positioning techniques to control the arrangement of elements on a webpage.

DAY 5

On the fifth day, I focused on responsive web design. I now know how to create layouts that work on a variety of screens and devices and about media queries. I studied how to make websites that are both mobile-friendly and accessible.

DAY 6

On the sixth day, I studied the area of typography in web design. I now know more about font families, styles, weights, and widths. I understood how to include web-safe fonts and use custom fonts using CSS. Additionally, I searched for methods to enhance readability through space and alignment.

DAY 7

I looked into the topic of color in web design on day seven. I learned about color models, such as RGB and hexadecimal numbers, and how to apply color to elements using CSS. I also considered color psychology and accessibility considerations in order to produce designs that are both visually beautiful and inclusive.

DAY 8

On the seventh day, I realized how important images are to the design of a website. I looked at techniques for reducing file size and making images more web-friendly. I also learned how to add images to HTML using the img element and how to position and style images using CSS.

DAY 9

I focused on the navigation design of web development on the ninth day. I learned about the several types of navigation menus, including vertical and horizontal menus, and how to create responsive and user-friendly navigation systems. I also came to see how important user-friendly navigation is to websites' usefulness.

DAY 10

On the ninth day, I began studying user experience (UX) design. I learned about UX principles and techniques for creating visually appealing, user-friendly websites that also matched the needs of their target market. I also investigated the use of wireframing and prototyping technologies in the planning and design of user-friendly interfaces.

DAY 11

On the eleventh day, I researched the concept of user interface (UI) design. I became familiar with UI components, such as buttons, forms, and icons, and I learned how to create and style them using HTML and CSS. I also researched UI design patterns and best practices to make visually stunning and user-friendly interfaces.

DAY 12

On the twelfth day, I learnt about online accessibility and its importance in inclusive web design. I knew the guidelines and procedures for creating websites that people with impairments could use and access. I also looked at tools for website evaluation and accessibility testing.

DAY 13

On the thirteenth day, my focus was on creating layouts for websites. I learned about grid layouts and how to create responsive grid layouts using CSS frameworks like Bootstrap. I have studied techniques for creating visually appealing and well-balanced page layouts.

DAY 14

On day fourteen, I started delving into the world of CSS frameworks. In order to speed up and streamline the web design process, I learned how to use pre-made CSS components and styles, as well as popular frameworks like Foundation and Bootstrap. I also investigated the tools for responsive design that these frameworks offer.

DAY 15

On the fifteenth day, I examined the concept of interaction design. I learned how to create engaging user experiences with HTML, CSS, and JavaScript, as well as interactive elements like sliders, forms, and buttons. I also looked for ways to add transitions and animations to enhance the user experience.

DAY 16

On the sixteenth day, I began studying web design optimization techniques. In an effort to increase website speed, I researched caching strategies, image optimization, and the minification of CSS and JavaScript files. I learned how to improve a website's overall performance and loading times for a better user experience.

DAY 17

On day seventeen, I began my JavaScript education. The fundamentals of JavaScript syntax, including variables, data types, operators, conditionals, and loops; functions and scope.

DAY 18

On the eighteenth day, I delved into the topic of responsive pictures in web design. I learned how to manage photographs differently based on the screen's quality and size. Event handling and DOM manipulation. Making websites that are interactive.

DAY 19

On the nineteenth day, I examined the idea of user-centered design, or UCD, in web development. I gained knowledge about conducting user research, getting user feedback, and putting iterative design methods into practice in order to create websites that cater to users' needs and preferences. I also knew that usability testing and user feedback were essential to making the design better.

DAY 20

On the twentieth day, I focused on web design trends and best practices. I researched minimalist, flat design, and tiny interactions as contemporary design trends. Additionally, I studied design principles including hierarchy, visual balance, and typographic choices in order to create visually appealing and modern website designs.

DAY 21

On the twenty-first day, I gained more knowledge about mobile-first design. I learned the importance of designing for mobile devices and how to create responsive designs that prioritize mobile consumers. I researched techniques including screen-size optimization, mobile-friendly design, and touch-friendly interfaces.

DAY 22

On the twenty-second day, I researched the concept of information architecture in web development. I acquired information about how to rationally and intuitively organize and structure website material using techniques like sitemaps, navigation menus, and classification. I also came to see how important it is to have a navigable website with an intelligible information structure.

DAY 23

On the twenty-third day, I focused on the role that branding and visual identity have in web design. I learned how to apply branding elements, such as logos, color schemes, and typography, to give a website a cohesive and identifiable visual identity. Additionally, I searched for strategies to harmonize the website design with the overall company image.

DAY 24

On the twenty-fourth day, I began studying about user testing and the usability of websites. I learned about the principles of usability as well as techniques for evaluating a website's usability, such as user testing, heatmaps, and analytics. I also investigated methods for gathering user feedback and making data-driven design decisions.

DAY 25

On day twenty-fifth day, I focused on web design for e-commerce. I learned how to create visually appealing product pages, expedite checkout processes, and foster trust and confidence for online shops. I researched ways to highlight products, include user reviews, and shorten the purchasing process.

DAY 26

On the twenty-sixth day, I examined the concept of search engine optimization, or SEO, in site design. I learned about search engine optimization techniques for websites that improve organic traffic and search engine rankings. I examined several aspects such as keyword research, on-page SEO, and link-building strategies.

DAY 27

I delved into the world of website statistics and tracking on day twenty-seven. In order to get insights into user behaviour and website performance, I learned how to configure and analyze data from website analytics packages like Google Analytics. I investigated techniques for using analytics data to make decisions about design that are driven by data.

DAY 28

On the twenty-eighth day, I focused on the importance of website security in web design. I now know the most effective techniques to defend websites from assaults and common security vulnerabilities. I investigated techniques such as configuring HTTPS secure protocols, preventing cross-site scripting (XSS), and conducting regular security audits.

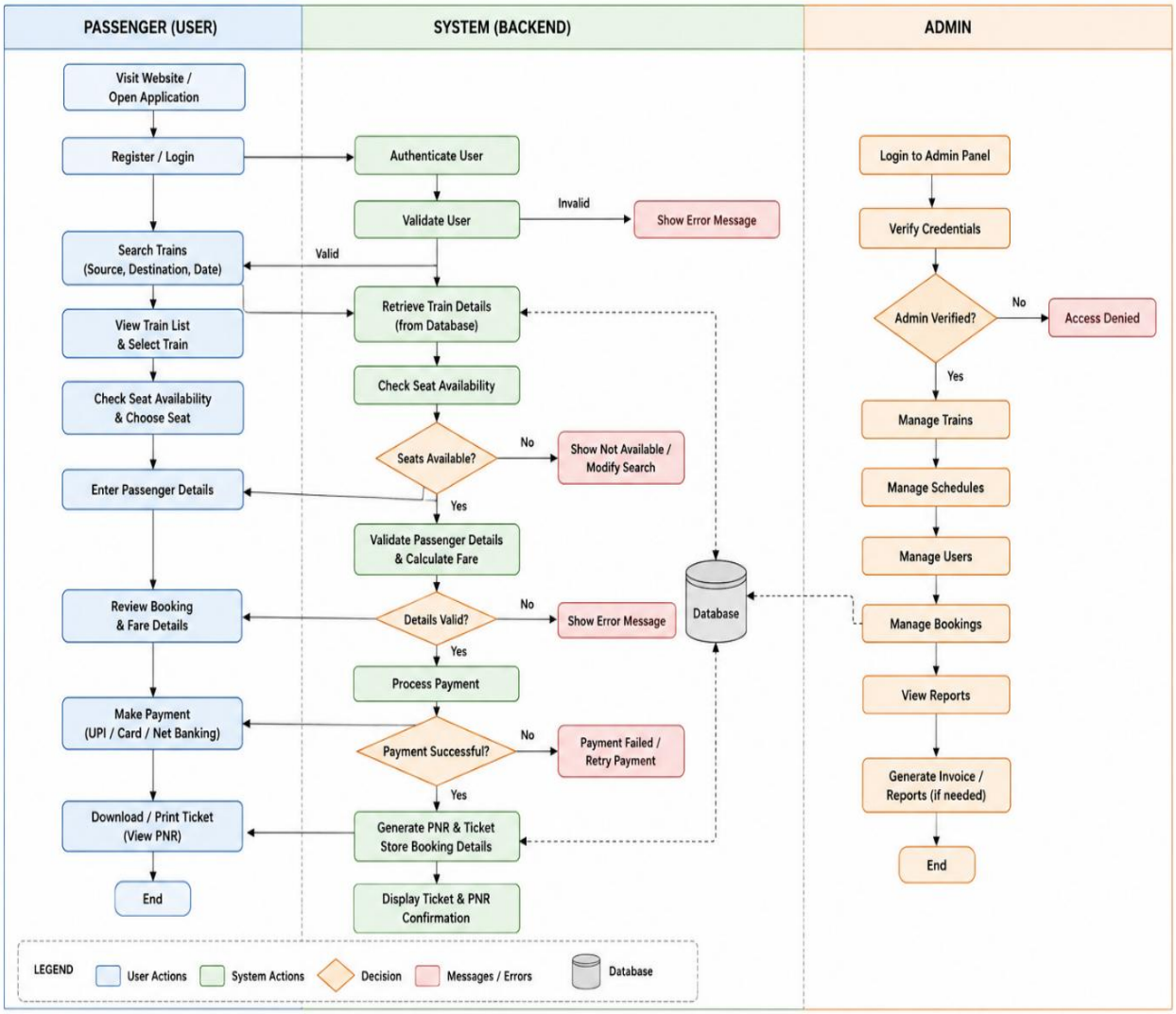
DAY 29

On the twenty-ninth day, I examined the concept of content management systems (CMS) in web design. I was given an explanation of popular content management systems (CMS) such as WordPress, Joomla, and Drupal, as well as an overview of their features for creating and managing content for websites. I investigated ways to improve and alter CMS functionalities.

DAY 30

I spent the day learning about database connectivity and Java Database Connectivity (JDBC). I gained knowledge on how to manage database transactions, run SQL queries, and connect applications to relational databases. I had practical expertise with the JDBC API to communicate with databases and carry out CRUD (Create, Read, Update, and Delete) operations.

4. BLOCK DIAGRAM



5. RAILWAY TICKET RESERVATION SYSTEM

The Railway Ticket Reservation System is a web-based application developed to simplify railway ticket booking and management. It allows passengers to register, search trains, check seat availability, book and cancel tickets, make payments, and view PNR details online.

The system provides separate functionalities for passengers and administrators. Passengers can manage their bookings, while administrators can manage trains, schedules, users, seats, and booking records.

The system uses a front-end, back-end, and database to process requests and store railway-related information. It aims to reduce manual effort, minimize booking errors, improve data management, and provide a fast, convenient, and user-friendly reservation experience.

5.1EXISTING SYSTEM

The existing railway ticket reservation process mainly involves online reservation platforms and, in some cases, manual booking through railway reservation counters. Passengers need to search for train details, check seat availability, provide passenger information, and complete the booking process through the available reservation facilities.

In the existing system, users may face difficulties such as complex navigation, longer processing time, limited accessibility, and difficulty in managing booking information**.** Manual reservation methods also require physical presence at reservation counters and may involve waiting in queues.

The management of train schedules, passenger details, seat availability, and booking records requires proper data handling. Errors or delays in updating information can affect the reservation process. Users may also need to access different sections of the system to check booking status, cancellation details, or PNR information.

Although existing railway reservation systems provide basic online booking facilities, there is a need for a simple, responsive, user-friendly, and well-organized web application that integrates major reservation activities into a single platform. The proposed system aims to address these limitations by providing an efficient web-based railway ticket reservation solution.

5.2PROPOSED SYSTEM

The proposed system is a web-based railway ticket reservation system designed to provide passengers with a simple, efficient, and convenient platform for railway ticket management. The system allows users to register and log in securely, search for trains, check seat availability, enter passenger details, and book tickets online.

The system integrates important reservation functions such as train search, seat selection, ticket booking, payment processing, ticket cancellation, and PNR status checking**.** After successful payment, the system generates a digital ticket containing passenger and journey details. Users can also view and manage their booking information through the system.

An administrator module is provided to manage train details, schedules, users, seat availability, and booking records. A centralized database stores and retrieves the required information efficiently. The system is designed with a responsive user interface so that it can be accessed through different devices.

The proposed system reduces manual effort, minimizes booking errors, improves data management, and provides faster access to railway reservation services. It also provides practical implementation of front-end development, back-end programming, database integration, authentication, and system testing**.**

5.3CAPABILITIES

The proposed railway ticket reservation system provides the following major capabilities:

- **User Registration and Login:** Allows passengers to create accounts and securely access the system.
- **Train Search:** Enables users to search trains based on source, destination, and journey date.
- **Seat Availability:** Displays available seats and allows passengers to select suitable seats.
- **Ticket Booking:** Enables passengers to enter their details and reserve tickets online.
- **Payment Processing:** Supports online payment for completing ticket reservations.
- **Digital Ticket Generation:** Generates a digital ticket with passenger, journey, and PNR details.
- **Ticket Cancellation:** Allows users to cancel confirmed bookings and manage their reservations.
- **PNR and Booking Status:** Enables passengers to view their booking and PNR information.
- **Admin Management:** Allows administrators to manage trains, schedules, users, seats, and bookings.
- **Database Management:** Stores passenger, train, booking, and payment information systematically.
- **Responsive Design:** Provides an accessible interface across different screen sizes.
- **Security and Validation:** Protects user information through authentication, validation, and controlled access.

6. PROPOSED METHOD

The proposed method describes the approach used to design and develop the Web-Based Railway Ticket Reservation System. The system is designed using a modular web application approach consisting of user authentication, train management, booking, payment, ticket generation, and administrative functions. The major modules of the proposed system are described below.

6.1 USER AUTHENTICATION AND REGISTRATION

The system provides a secure registration and login facility for passengers and administrators. New users can create an account by providing the required details, while registered users can log in using their credentials. Authentication helps protect user information and ensures that only authorized users can access specific system functions.

6.2 TRAIN SEARCH AND SEAT AVAILABILITY

Passengers can search for trains by entering details such as source, destination, and journey date. The system retrieves matching train information from the database and displays available trains, schedules, and seat availability. This helps passengers select a suitable train before proceeding with the booking process.

6.3 TICKET BOOKING

After selecting a train, passengers can enter the required passenger details and select available seats. The system validates the entered information and calculates the applicable fare. Once the booking is confirmed, the reservation details are stored in the database.

6.4 PAYMENT AND TICKET GENERATION

The system provides a payment facility for completing the reservation process. After successful payment, the booking is confirmed and a unique PNR number is generated. A digital ticket containing passenger, train, journey, seat, and PNR details is then generated for the passenger.

6.5 TICKET CANCELLATION AND BOOKING STATUS

Passengers can view their booking details and cancel tickets when required. The system updates the booking and seat availability information after cancellation. Users can also check their PNR and booking status through the application.

6.6 ADMINISTRATIVE MANAGEMENT

The administrator can manage important railway information through the admin module. This includes adding, updating, and deleting train details, managing schedules, monitoring users, and viewing booking records. The administrator helps maintain accurate and up-to-date system information.

6.7 DATABASE MANAGEMENT

A centralized database is used to store and manage user, train, passenger, seat, booking, payment, and PNR information. Proper database management ensures data consistency, quick retrieval, and reliable storage of reservation records.

6.8 SYSTEM INTEGRATION AND TESTING

All modules are integrated to form a complete web-based railway reservation system. The application is tested to identify and correct errors and to ensure that major functions such as login, train search, booking, payment, cancellation, and ticket generation work correctly. The system is also evaluated for usability, reliability, security, and performance.

7. ALGORITHMS

7.1 USER AUTHENTICATION ALGORITHM

Input: Username/Email and Password
Output: Successful Login or Error Message

Steps:

1. Start the process.
2. Enter username/email and password.
3. Validate the entered details.

4. Check the credentials in the database.
5. If the credentials are valid, allow the user to access the system.
6. If the credentials are invalid, display an error message.
7. End the process.

7.2 TRAIN SEARCH ALGORITHM

Input: Source, Destination, and Journey Date

Output: List of Available Trains

Steps:

1. Start the process.
2. Enter source, destination, and journey date.
3. Validate the search details.
4. Retrieve matching train records from the database.
5. Check train availability for the selected date.
6. Display the available trains and schedules.
7. If no train is available, display an appropriate message.
8. End the process.

7.3 SEAT AVAILABILITY ALGORITHM

Input: Selected Train and Journey Date

Output: Available Seats

Steps:

1. Select a train and journey date.
2. Retrieve the seat information from the database.
3. Check the current booking status of the seats.
4. Identify available and occupied seats.
5. Display the available seats to the passenger.
6. Allow the passenger to select an available seat.
7. End the process.

7.4 TICKET BOOKING ALGORITHM

Input: Train, Seat, and Passenger Details

Output: Booking Confirmation

Steps:

1. Start the booking process.
2. Select the train and available seat.
3. Enter passenger details.
4. Validate the passenger information.
5. Calculate the ticket fare.
6. Proceed to payment.
7. If payment is successful, store the booking details.
8. Update the selected seat as occupied.
9. Generate booking confirmation and PNR.
10. End the process.

7.5 PAYMENT PROCESSING ALGORITHM

Input: Booking Details and Payment Information

Output: Payment Status

Steps:

1. Start the payment process.
2. Display the calculated ticket fare.
3. Enter the required payment details.
4. Validate the payment information.
5. Process the payment.
6. If payment is successful, confirm the booking.
7. If payment fails, display an error message and allow the user to retry.
8. Store the payment status in the database.
9. End the process.

7.6 PNR AND TICKET GENERATION ALGORITHM

Input: Confirmed Booking Details

Output: PNR and Digital Ticket

Steps:

1. Retrieve the confirmed booking details.
2. Generate a unique PNR number.
3. Store the PNR with the booking record.
4. Prepare the digital ticket containing passenger, train, journey, seat, and fare details.
5. Display the ticket to the passenger.
6. Allow the passenger to view or download the ticket.
7. End the process.

7.7 TICKET CANCELLATION ALGORITHM

Input: PNR / Booking ID

Output: Cancellation Confirmation

Steps:

1. Enter the PNR or booking ID.
2. Retrieve the booking details.
3. Verify that the booking exists and can be cancelled.
4. Confirm the cancellation request.
5. Update the booking status as cancelled.
6. Release the reserved seat.
7. Update the seat availability in the database.
8. Process the applicable refund, if required.
9. Display the cancellation confirmation.
10. End the process.

8. CODE IMPLEMENTATION

8.1 Front-End Implementation

Example – Train Search Form

```
<!DOCTYPE html>
<html>
<head>
    <title>Train Search</title>
</head>
<body>
<h2>Search Trains</h2>
<form action="/search" method="POST">
    <label>From:</label>
    <input type="text" name="source" required>
   <label>To:</label>
    <input type="text" name="destination" required>
   <label>Journey Date:</label>
    <input type="date" name="date" required>
  <button type="submit">Search Train</button>
</form>
</body>
</html>
```

8.2 User Registration and Login

Example – Registration Form

```
<form action="/register" method="POST">

    <input type="text"
           name="name"
           placeholder="Enter Name"
           required>
    <input type="email"
           name="email"
           placeholder="Enter Email"
           required>
    <input type="password"
           name="password"
           placeholder="Enter Password"
           required>
    <button type="submit">Register</button>
</form>
```

Example – Login Route

```
app.post('/login', (req, res) => {
  const { email, password } = req.body;
  const sql =
    "SELECT * FROM users WHERE email = ? AND password = ?";
  db.query(sql, [email, password], (err, result) => {
    if (err) {
      return res.status(500).send("Database Error");
```

```
        }
        if (result.length > 0) {
            res.send("Login Successful");
        } else {
            res.send("Invalid Email or Password");
        }
    });
});
```

8.3 Train Search and Seat Availability

```
app.post('/search', (req, res) => {
    const { source, destination, date } = req.body;
    const sql = `
        SELECT * FROM trains
        WHERE source = ?
        AND destination = ?
        AND journey_date = ?
    `;
    db.query(
        sql,
        [source, destination, date],
        (err, result) => {
            if (err) {
                return res.status(500).send("Database Error");
            }
```

```
            res.json(result);
}
    );
});
```

Example database table

```
CREATE TABLE trains (
    train_id INT PRIMARY KEY AUTO_INCREMENT,
    train_name VARCHAR(100),
    source VARCHAR(50),
    destination VARCHAR(50),
    journey_date DATE,
    available_seats INT
);
```

8.4 Ticket Booking

```
app.post('/book', (req, res) => {
    const {
        user_id,
        train_id,
        passenger_name,
        seat_no
    } = req.body;
    const sql = `
        INSERT INTO bookings
        (user_id, train_id, passenger_name, seat_no, status)
```

```
        VALUES (?, ?, ?, ?, 'Confirmed')
    `;
    db.query(
        sql,
        [user_id, train_id, passenger_name, seat_no],
        (err, result) => {
            if (err) {
                return res.status(500)
                    .send("Booking Failed");
            }
            res.send("Ticket Booked Successfully");
        }
    );
});
```

8.5 Payment Processing

```
app.post('/payment', (req, res) => {
    const { booking_id, amount } = req.body;
    const sql = `
        INSERT INTO payments
        (booking_id, amount, status)
        VALUES (?, ?, 'Successful')
    `;
    db.query(
        sql,
```

```
        [booking_id, amount],
        (err, result) => {
            if (err) {
                return res.status(500)
                    .send("Payment Failed");
            }
            res.send("Payment Successful");
        }
    );
});
```

Payment table

```
CREATE TABLE payments (
    payment_id INT PRIMARY KEY AUTO_INCREMENT,
    booking_id INT,
    amount DECIMAL(10,2),
    status VARCHAR(20)
);
```

8.6 PNR and Digital Ticket Generation

```
function generatePNR() {
    return "PNR" +
           Math.floor(100000 + Math.random() * 900000);
}
const pnr = generatePNR();
console.log("Generated PNR:", pnr);
```

SQL

```
UPDATE bookings
SET pnr = 'PNR583214',
   status = 'Confirmed'
WHERE booking_id = 101;
```

8.7 Ticket Cancellation

```
app.post('/cancel', (req, res) => {
   const { booking_id } = req.body;
   const sql = `
      UPDATE bookings
      SET status = 'Cancelled'
      WHERE booking_id = ?
   `;
   db.query(sql, [booking_id], (err, result) => {
       if (err) {
          return res.status(500)
              .send("Cancellation Failed");
       }
       res.send("Ticket Cancelled Successfully");
   });
});
```

8.8 Database Connectivity

```
const mysql = require('mysql2');
```

```
const db = mysql.createConnection({
    host: 'localhost',
    user: 'root',
    password: 'your_password',
    database: 'railway'
});
db.connect((err) => {
    if (err) {
        console.log("Database Connection Failed");
    } else {
        console.log("Database Connected Successfully");
    }
});
```

8.9 Admin Module

```
app.get('/admin/trains', (req, res) => {
    const sql = "SELECT * FROM trains";
    db.query(sql, (err, result) => {
        if (err) {
            return res.status(500)
                .send("Database Error");
        }
        res.json(result);
    });
```

});

8.10 System Integration

Passenger

↓

Web Interface

↓

HTML / CSS / JavaScript

↓

Node.js + Express.js

↓

Application Logic

↓

MySQL Database

↓

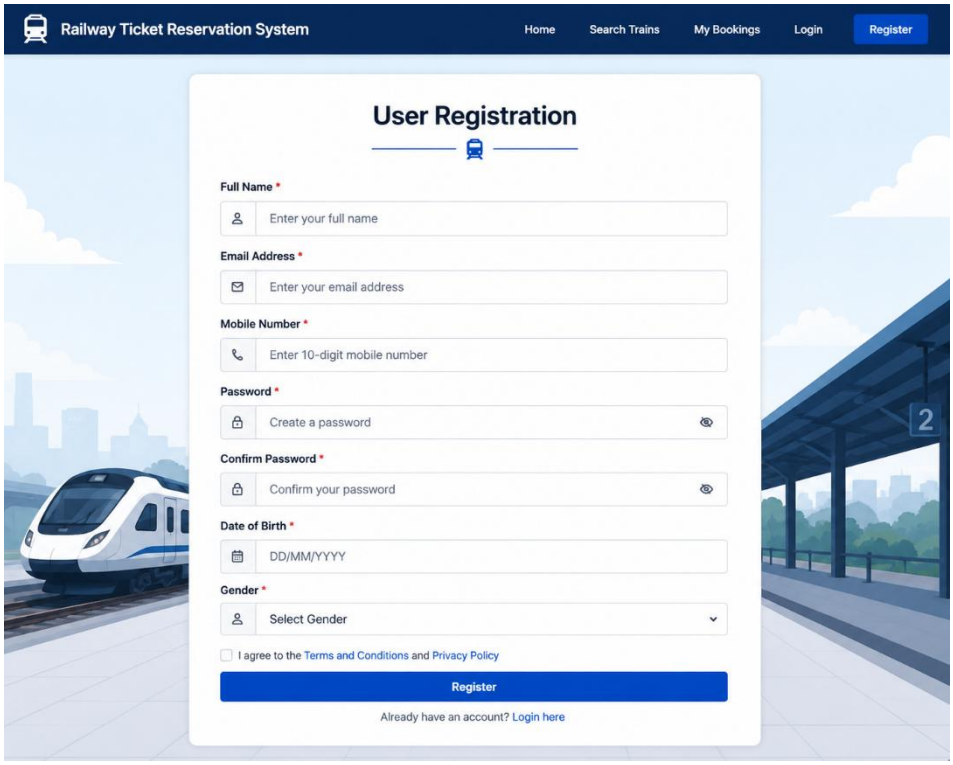
Booking / Payment / PNR

↓

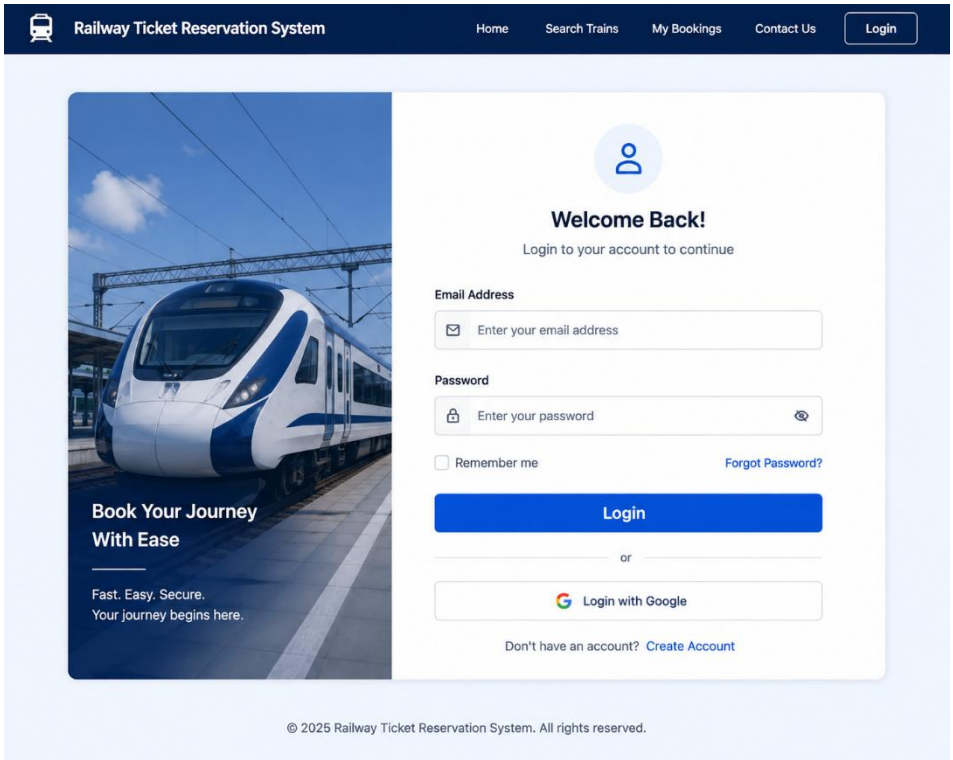
Digital Ticket

9. SAMPLE OUTPUT

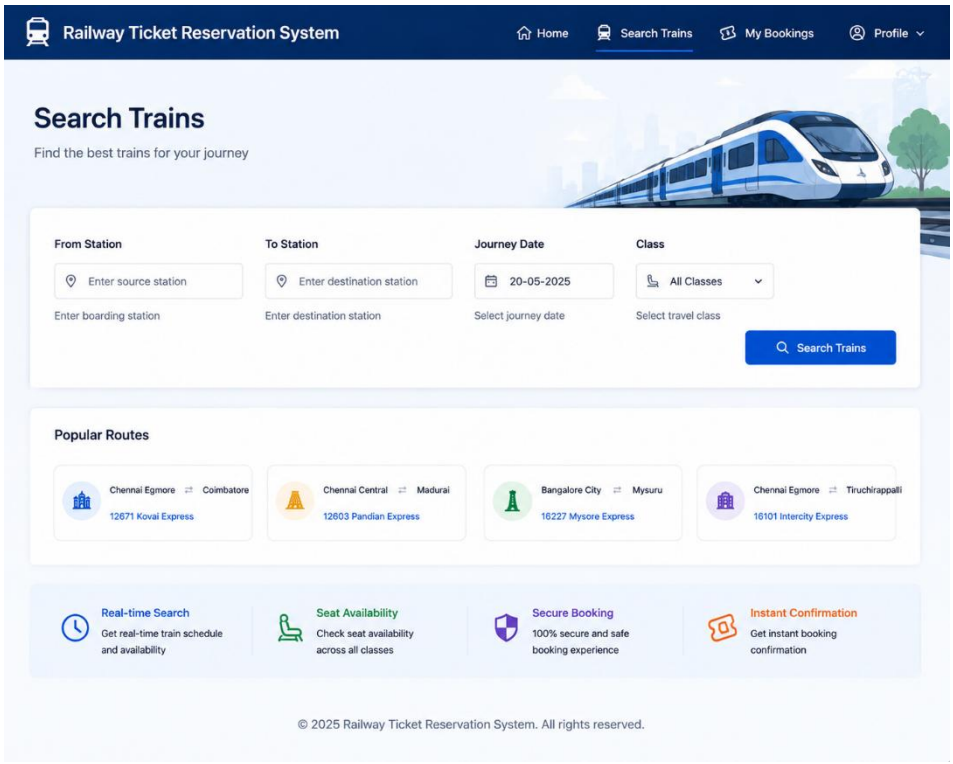
9.1 USER REGISTRATION PAGE



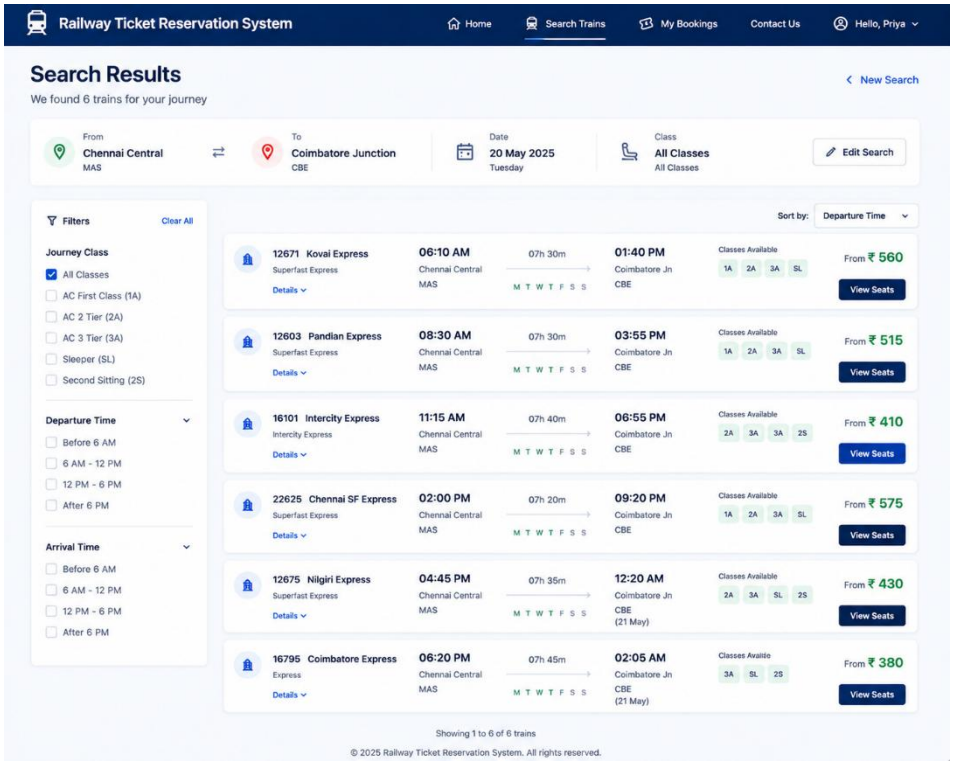
9.2 USER LOGIN PAGE



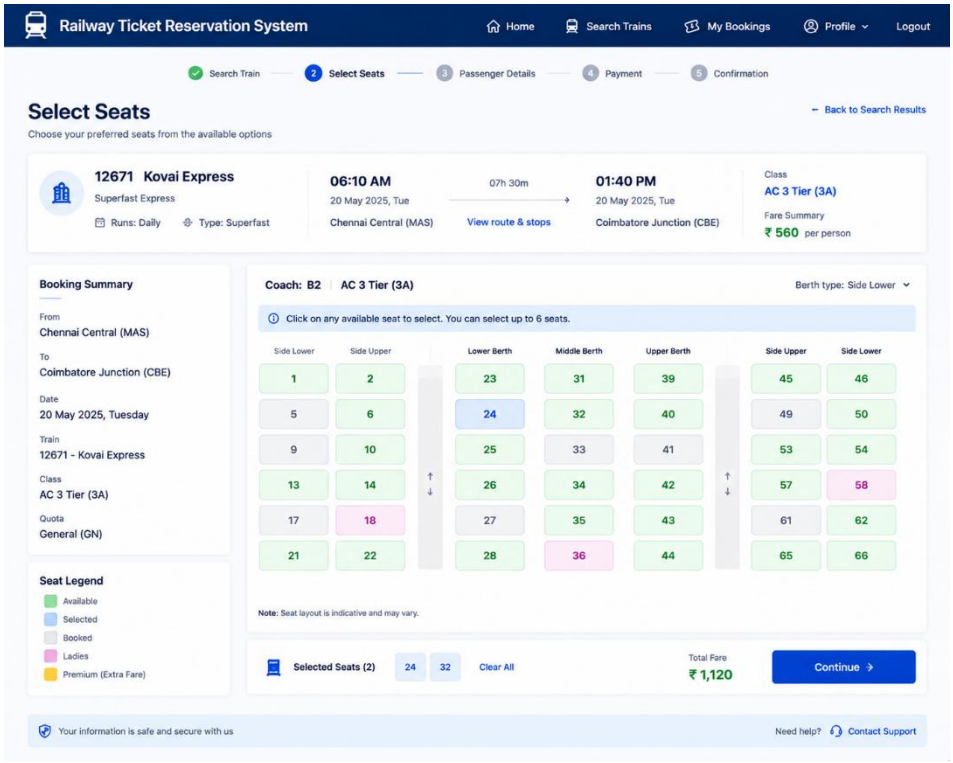
9. 3 TRAIN SEARCH PAGE



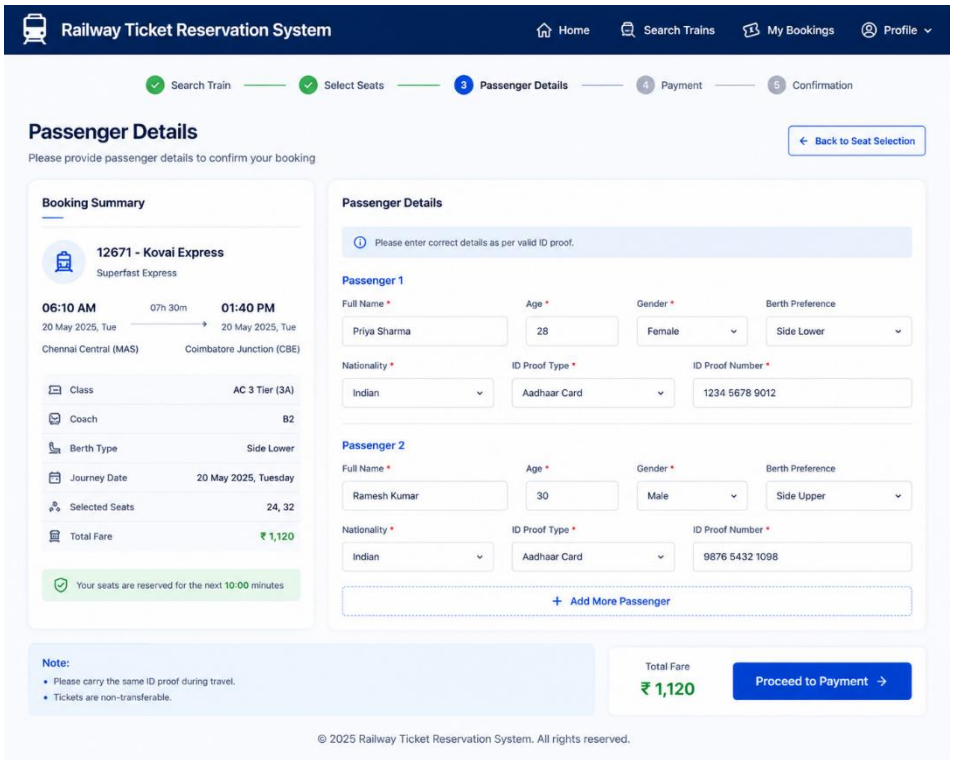
9.4 TRAIN SEARCH RESEARCH PAGE



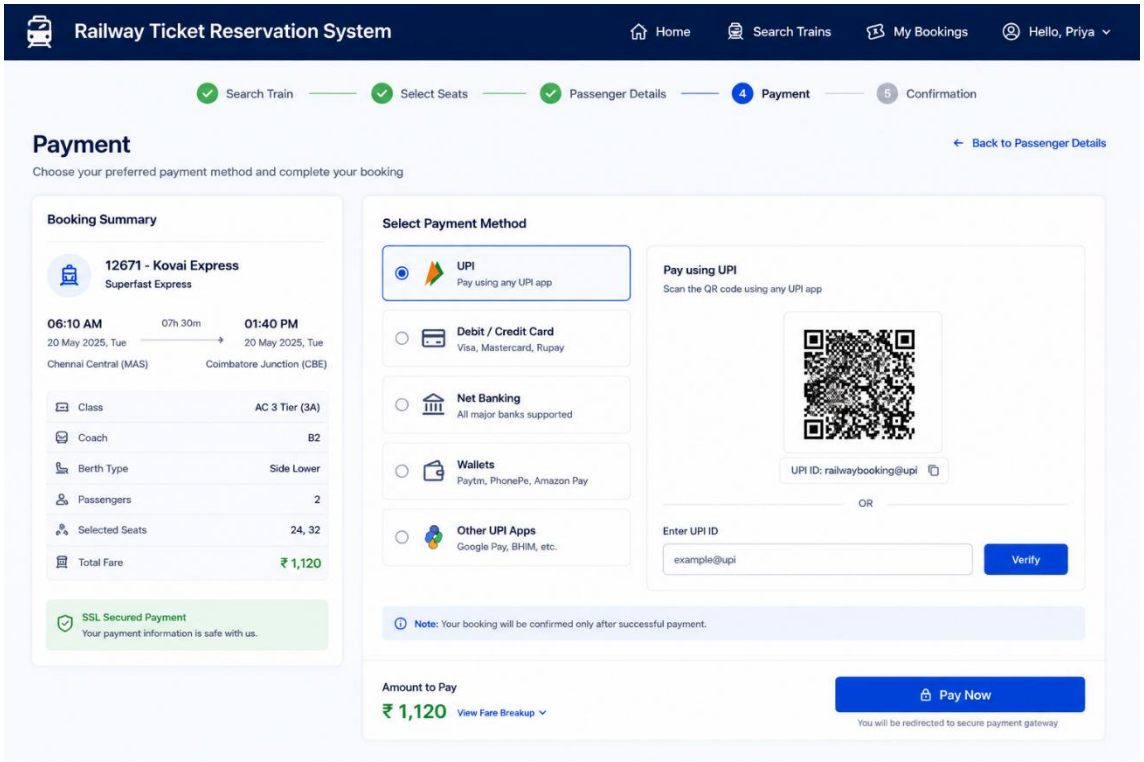
9.5 SEAT AVAILABILITY /SEAT SELECTION PAGE



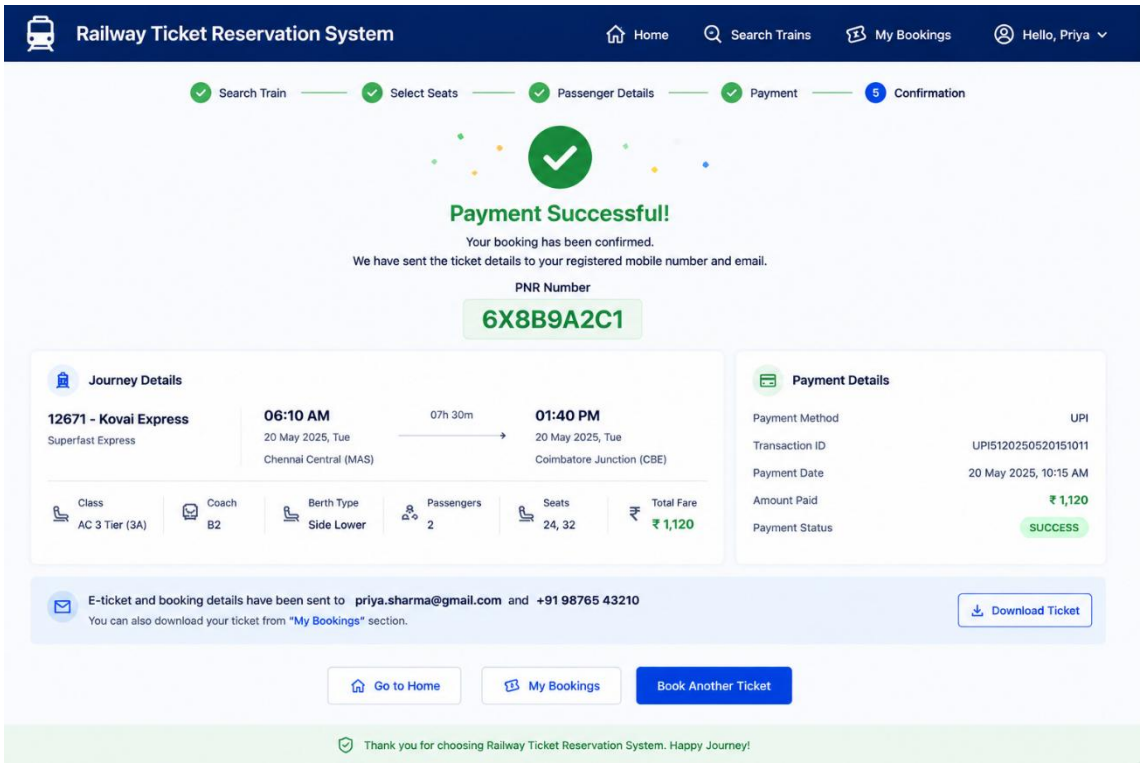
9.6 PASSENGER DETAILS/BOOKING CONFIRMATION PAGE



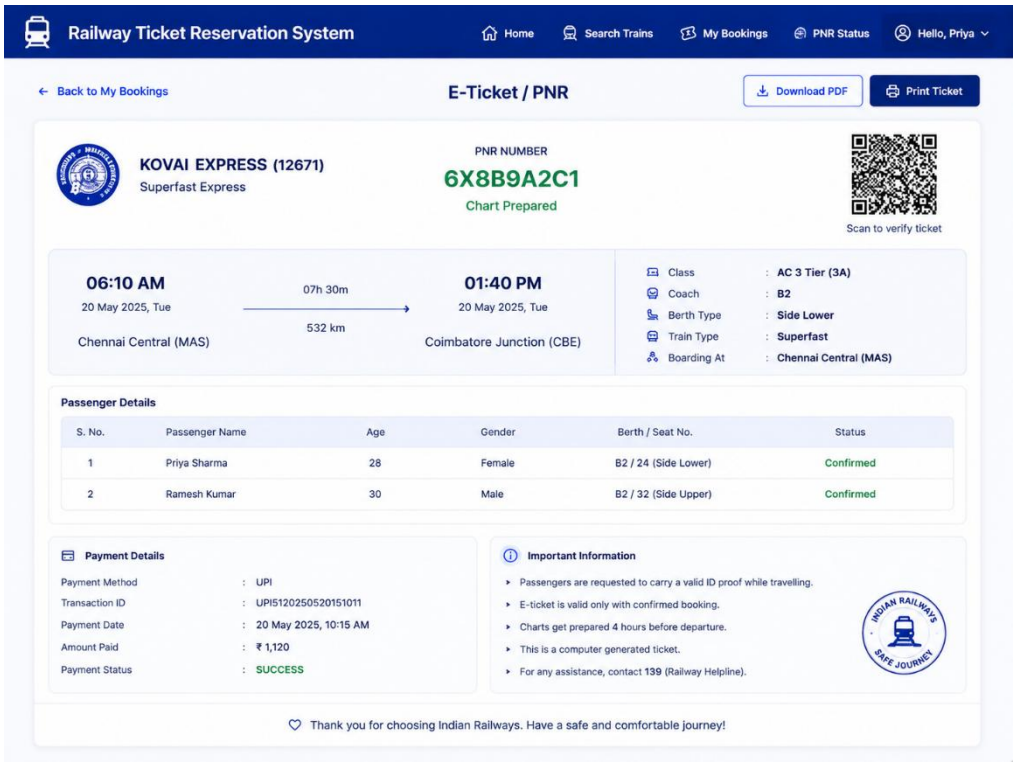
9.7 PAYMENT METHOD PAGE



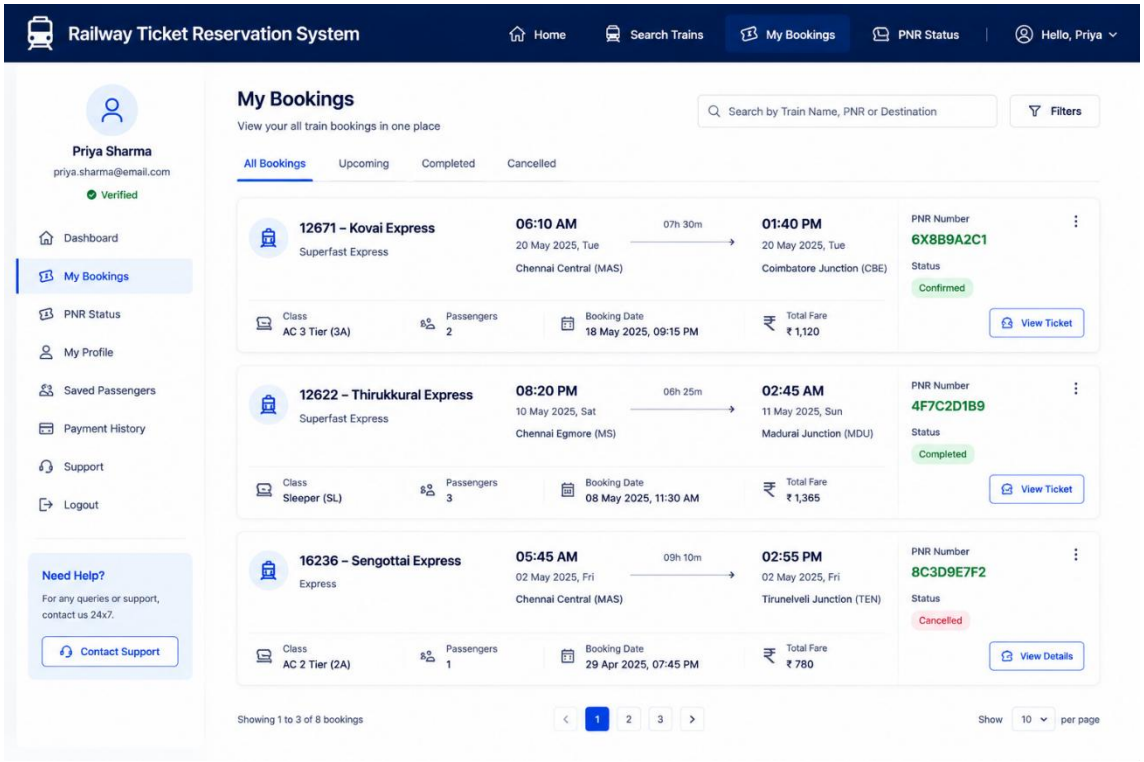
9.8 PAYMENT SUCCESSFUL PAGE



9.9 DIGITAL RAILWAY TICKET /PNR PAGE



9.10 BOOKING HISTORY/MY BOOKINGS PAGE



10. ANALYSIS

The Web-Based Railway Ticket Reservation System was analyzed to evaluate its functionality, usability, and efficiency. The system provides important features such as user registration, login, train search, seat availability, ticket booking, payment, PNR generation, cancellation, and booking history.

The centralized database efficiently manages user, train, passenger, booking, seat, and payment information. Authentication and validation help maintain data security and accuracy.

The system provides a simple, responsive, and user-friendly interface, allowing passengers to complete the reservation process easily. The admin module helps manage trains, schedules, users, and bookings efficiently.

Testing of major functions such as login, train search, seat selection, booking, payment, ticket generation, and cancellation helps ensure reliable system performance.

Overall, the analysis shows that the proposed system can reduce manual effort, minimize errors, save processing time, and improve the overall railway ticket reservation experience.

11. CONCLUSION

The “Design and Development of a Web-Based Railway Ticket Reservation System” was successfully developed to provide a simple, efficient, and user-friendly solution for railway ticket reservation. The main objective of the project was to reduce the difficulties involved in traditional reservation procedures and provide passengers with an easy platform to access railway services online.

The system provides important functionalities such as user registration and login, train search, seat availability checking, passenger details management, ticket booking, payment processing, ticket cancellation, PNR generation, digital ticket generation, and booking history. These features are integrated into a single platform, allowing passengers to complete the reservation process in a systematic and convenient manner.

The system also includes an administrator module for managing train information, schedules, users, seat availability, and booking records. A centralized database helps store and retrieve important information efficiently, while authentication and validation mechanisms help maintain data accuracy and controlled access.

The project follows the major stages of web application development, including requirement analysis, system design, front-end development, back-end development, database integration, testing, and debugging. The responsive interface provides easy navigation and allows users to access the system through different devices.

The development of this project provided valuable practical experience in HTML, CSS, JavaScript, server-side programming, database management, authentication, application integration, testing, and debugging. It helped in understanding how different components of a web application work together to create a complete real-world solution.

Overall, the Web-Based Railway Ticket Reservation System provides a convenient and organized approach to railway ticket management. The system helps reduce manual effort, minimize booking errors, save processing time, and improve reservation data management. With future enhancements such as real-time train tracking, secure payment gateway integration, OTP authentication, mobile application support, and notifications, the system can be further developed into a more advanced and scalable railway reservation platform.

12. FUTURE WORK

The system can be further improved by adding advanced features and technologies. Future enhancements may include:

- Integration with real-time railway schedules and live train status.
- Integration with secure and real-world online payment gateways.
- Addition of email and SMS notifications for booking and cancellation.
- Implementation of OTP-based authentication for improved security.
- Development of a mobile application for Android and iOS.
- Addition of AI-based train and travel recommendations.
- Integration of real-time seat availability and automatic seat allocation.
- Addition of multilingual support for better accessibility.
- Implementation of advanced admin reports and analytics.
- Deployment of the application on a cloud platform for scalability and accessibility.

13. REFERENCES

You can use the following references in your internship report:

1. Jon Duckett, HTML and CSS: Design and Build Websites, Wiley.
2. Jon Duckett, JavaScript and JQuery: Interactive Front-End Web Development, Wiley.
3. Ethan Brown, Web Development with Node and Express, O'Reilly Media.
4. Robert C. Martin, Clean Code: A Handbook of Agile Software Craftsmanship, Prentice Hall.
5. MySQL Documentation, MySQL Reference Manual.
6. Node.js Documentation, Node.js Official Documentation.
7. Express.js Documentation, Express.js Official Documentation.
8. MDN Web Docs, HTML, CSS, and JavaScript Web Development Documentation.
9. W3Schools, Web Development Tutorials and References.
10. GeeksforGeeks, Web Development and Database Management References.

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SOFTWARE DEVELOPMENT AND TRAINING

Certificate of Accomplishment

This is certify that Mr./Miss. MADHUMITHA R (810023205064)

Student of UNIVERSITY COLLEGE OF ENGINEERING BIT CAMPUS ANNA UNIVERSITY - TRICHY

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